
The Problem with Affordance

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ABSTRACT This article reviews the concept of ‘affordance’, a term widely used in the literature on learning and technology to try and explain the properties technologies have. It is argued that the concept has drifted so far from its origins that it is now too ambiguous to be analytically valuable. In addition, it is suggested that its origins in ecological perception – where relatively primitive claims are made at the level of species about animal-object relationships, framed in an evolutionary timescale – have little direct relevance to the moment-to-moment interactions of an individual using a specific artefact. Because of this, it is proposed that an alternative way of talking about technology use is required. A stylistic analysis of technologies, drawing on Bakhtin’s analysis of novels, is outlined as one potential alternative.

Introduction

The concept of ‘affordance’ features frequently in research into technology and learning. The term is used, in various ways, to explain people’s relationships with technologies. (A fuller review of these uses will be provided later in the article.) Unsurprisingly, this makes the concept extremely important for this field – potentially a foundational concept for research (Conole & Dyke, 2004a). This proposal has been challenged (Boyle & Cook, 2004), but primarily at an operational level – more critical questions about the concept, such as those asked in the field of human–computer interaction (e.g. Albrechtsen et al, 2001), remain unasked here.

The purpose of this article is to take a critical position by exploring the history of affordance, the assumptions underlying it and implications of using it. This will be achieved by tracing the discourse of affordance from its inception in Gibson’s theory of ecological perception, through the literature on design, into human–computer interaction and finally to learning and technology.

A History of the Discourse

The Origin of Affordances

Gibson’s book *The Ecological Approach to Visual Perception* (1979) grew from studies during World War II into the depth perception of pilots. Frustrated that these failed to predict flight performance, he devised an alternative theory of perception, proposing an approach that, initially, appears sympathetic to current situated perspectives on action:

When no constraints are put on the visual system, we look around, walk up to something interesting and move around it so as to see it from all sides, and go from one vista to another.

That is natural vision, and that is what this book is about. (Gibson, 1979, p. 1)

The theory of ecological perception proceeds by specifying a unit of analysis (‘The words *animal* and *environment* make an inseparable pair. Each term implies the other’, [p. 8]), and concepts are defined (e.g. mediums, substances and the surfaces that separate them, [p. 16]) that permit an ecological epistemology to be developed: ‘Because of illumination the animal can see things;

because of sound it can hear things; because of diffusion it can smell things. The medium thus contains information about things that reflect light, vibrate or are volatile' (p. 17).

This notion that information is simply present in the 'ambient optic array' (the convergence, at our point of perception, of various diffracted rays of light) is thus used to challenge common theories of meaning:

The ambient stimulus information available in the sea of energy around us ... is not transmitted, does not consist of signals, and does not entail a sender and receiver. The environment does not communicate with the observers who inhabit it. Why should the world speak to us? ... The world is *specified* in the structure of the light that reaches us, but it is entirely up to us to perceive it. (p. 63)

It is this spirit of transparent information that underlies the concept of affordance: 'The *affordances* of the environment are what it *offers* the animal, what it *provides* or *furnishes*, either for good or ill' (p. 115, original emphases). Gibson, emphasising the 'complementarity of the animal and environment', stresses that affordances are '*relative to the animal*' (p. 115). This has interesting consequences:

The affordances of the environment ... are in a sense objective, real and physical. ... An affordance is neither an objective property nor a subjective property; or it is both if you like. An affordance cuts across the dichotomy of subjective-objective and helps us to understand its inadequacy. It is equally a fact of the environment and a fact of behavior. It is both physical and psychical, yet, neither. An affordance points both ways, to the environment and to the observer. (p. 129; cf. p. 41)

This claim is partly explained by his epistemological commitments. Specifically and repeatedly, he attacks the idea that mental activity is involved in perception. ('These are still the operations of the mind upon the deliverances of the sense, and there are too many perplexities entailed in this theory. It will not do, and the approach should be abandoned' [p. 238; cf. p. xiii, 2].) He dismisses the Paigetian approach that 'emphasizes what the children believe instead of what they see' (p. 195) and adopts a materialist, essentialist position: 'Let us suppose that a kind of essential structure underlies the superficial structure of an array when the point of observation moves. This essential structure consists of what is invariant despite the change' (p. 73).

Given this position, how should Gibson's claim that affordances are neither subjective nor objective be interpreted? This claim begs the question of what the relationship consists of. There seem to be few possibilities, and none of them are satisfactory. An initial position might be that the relationship consists of all the interactions between object and perceiver that are possible. What, though, is the ontological status of this set of actions? 'Possible' leaves nothing substantial to work with, unless a definitive list of possibilities can be constructed from the properties of each element separately, and their various interactions. This is only feasible if people and things can be exhaustively described, which presupposes both an essentialist, positivist epistemology (which is consistent with Gibson's position) and a finite number of characteristics (which is not: see p. 243). Lacking this, 'possible' is a work of imagination and 'affordance' becomes speculative rather than analytic. This possibility will be returned to later.

If the nature of this relationship is not speculative, it must be either historical or normative. If it were historical, all that could be said then is that a thing *afforded* something to someone in a specific circumstance, not that it *affords* it in general. This path of diminishing returns trivialises the concept analytically. (Chemero [2003] offers this as a solution, but attempts to preserve generality by arguing that affordances depend upon potential, not actual, observers – inconsistently reverting to 'possibilities'.)

Next are normative interpretations (which Chemero sees no problem with), implying things about 'typical' or 'expected' interactions – however, 'typical' presupposes an historical analysis, and 'expected' implies a mental activity of the kind that Gibson wishes to rule out. It also fails to specify what counts as 'normal' (Potter & Wetherell, 1987).

This suggests that, in spite of the claimed relational basis for affordances, they are most coherently interpreted either as works of imagination or as positivistic properties of objects, albeit ones activated or perceived by people and animals. They arise from the environment, where the 'invariants' that specify them are located. This interpretation is consistent with Gibson's own

speculations: 'Perhaps the composition and layout of surfaces *constitute* what they afford' (Gibson, 1979, p. 127).

Ecological psychologists recognise this problem – for example, Chemero (2003, pp. 182-183) referred to affordances as 'impossible, ghostly entities, entities that no respectable scientist ... could have as part of their ontology'.

It is interesting to consider the kinds of claims Gibson makes for affordances. Most are animalistic or primitive (Gibson, 1971); technology, if mentioned, is stone-age (e.g. clubs, hand axes, spears, awls [Gibson, 1979, p. 40; cf. p. 128]): 'All these offerings of nature, these possibilities or opportunities, these *affordances* as I will call them, are invariant. They have been strikingly constant throughout the whole evolution of animal life' (1979, p. 19).

Gibson does attempt to extend this theory to more complex affordances – but the examples are sparse and unconvincing (e.g. 1979, p. 42), resorting to what can only be interpreted as visual stereotypes (e.g. 'Mates, and amiable animals are distinguished from ... hostile animals by their shapes, colors, textures, and deformations' [1979, p. 232]).

He also concedes that 'information pickup' (direct, unprocessed perception) is not always straightforward (e.g. 1979, pp. 152, 157) and that 'misinformation' may result in 'misperception' (1979, pp. 142, 243). He attempts to explain away this problem by shifting responsibility onto the inattentive perceiver, rather than his positivistic epistemology or proposed perceptual system (1979, p. 243). Perceptual problems relating to dreams, drugs or delusions are also the perceiver's fault, he argues, since they cannot or will not 'make the ordinary tests for reality' (1979, p. 257). The problems of information pickup thus remain unresolved, practically. This renders affordances problematic. Again, Gibson can only offer visual stereotypes as a solution (with no mention of how these are acquired).

The misperceiving of affordances is a serious matter. As I noted in Chapter 8, a wildcat may look like a cat. (But *does* he look just like a cat?) A malevolent man may act like a benevolent one. (But *does* he exactly?) The line between the pickup of information and the failure to pick up information is hard to draw. (1979, p. 244)

This analysis is not reassuring for educationalists; nor is the way Gibson talks about information, knowledge and learning: 'The term *information* cannot have its familiar dictionary meaning of *knowledge communicated to a receiver*. This is unfortunate, and I would use another term if I could' (1979, p. 242).

He explicitly discounts the kinds of knowledge educationalists would value, focusing almost exclusively on 'knowledge of the environment', rather than 'knowledge that comes from parents, teachers, pictures and books. ... This is a different kind of knowledge' (1979, p. 253). As soon as affordances are introduced, Gibson appropriates 'value' and 'meaning' in a similar way (1979, p. 140) – as environmental properties that can be 'directly perceived' (1979, p. 127). This, he alleges, does away with the need for learning (see, for example, 1979, pp. 128, 40). Gibson tries to explain other sources of knowledge, such as writing and art:

Pictures and sculptures ... *contain* information and make it available for anyone who looks. They nevertheless are, like the spoken and written words of language, *man-made*. They provide information that ... is mediated by the perception of the first observer. They do not permit firsthand experience – only experience at second hand. (1979, p. 63; cf. p. 42)

Elsewhere he attempts to explain how indirect knowledge differs from 'normal' ecological knowledge, differentiating direct perception from mediated knowledge. The latter is still directly perceived (it still acts as a stimulus), but operates differently; he proposes that 'perceptual mediators' are invented and have to be learnt (Gibson, 1965), but he does not explain how or provide an account of meaning in relation to mediation. He also does not explain any relationship between this and affordance.

Indeed, in spite of his attempts to argue the irrelevance of the mind, Gibson repeatedly states that affordances need learning (e.g. 1979, pp. 143, 194, 235; Gibson, 1971). Even for the most 'basic' affordances, he concedes that perception might need to develop in some way – something that most people would call a kind of learning even if Gibson avoids describing it as such (Gibson, 1979, pp. 168, 241, 253).

Gibson's defence involves denying, again, the role of the mental, arguing instead for a process of 'attunement':

The theory of information pickup ... needs to explain learning, *that is*, the improvement of perceiving with practice and the education of attention, but not by an appeal to the catch-all of past experience or to the muddle of memory. The state of a perceptual system is altered when it is attuned to information of a certain sort. The system has become sensitized. (Gibson, 1979, p. 254; emphasis added)

Unsurprisingly (given his radical agenda), other theorists of perception have challenged Gibson's position. Gregory, for example (1998), presents a position far easier to reconcile with constructivist perspectives.

Perceptions are predictive hypotheses, based on knowledge stored from the past. ... Perceptions are not regarded as internal pictures or sounds, but rather as language-like descriptions coded, we suppose, by brain structures of what may be out there. We carry in our heads predictive hypotheses of the external world of objects and of ourselves. (Gregory, 1998, p. 1693)

Gregory's position is not actually as far from Gibson's as might at first be supposed:

The perceptual process is not primarily one of supplementing or contracting but of *selecting*. If so, the contribution of the perceiver is only one of paying attention, of looking, of exploring, of adjusting the eyes, and of detecting the 'formless invariants'. (Gibson, 1965)

The crucial difference is that Gibson fails to account for how this selecting takes place; the intentionality of selection, recognised by Gregory as knowledge-informed, remains mysterious in Gibson's scheme.

Other commentators have attempted to redeem Gibson's position through clarification.

The existence of the affordance is independent of the actor's experiences and culture, whereas the ability to perceive the affordance may be dependent on these. Thus, an actor may need to learn to discriminate the information in order to perceive directly. In this way learning can be seen as a process of discriminating patterns in the world, rather than one of supplementing sensory information with past experience. (McGrenere & Ho, 2000, p. 2)

But this does not resolve the situation. This distinction contradicts Gibson's assertion that perception is 'direct' and yet still runs into the problem of what the 'relationship' between actor and environment consists of. It also renders affordances unknowable – a problem that will be returned to in the next section.

To summarise, Gibson advanced a radical agenda for perceptual psychology – one that is situated, but anti-'mentalist'. Affordances are positioned as relational, but Gibson failed to explain what this relationship consists of. Although he tries to argue that perception is direct, Gibson cannot avoid talking about learning (sometimes under pseudonyms). Whether or not affordances exist, perceiving them is problematic – they are relevant ontologically, but not epistemologically. Although the idea of affordance might be explicable as a normative account, and thus acceptable at the ecological level of species (e.g. justified through 'natural selection'; Chemero, 2003), this is of little comfort to educators, concerned with specific people or groups learning things.

Norman and the Adoption of Affordances for Design

Like Gibson, Norman embarked on a radical and passionate project – *The Psychology of Everyday Things* (POET; Norman, 1988) – because of personal frustration; in this case, with poor designs (Norman, 1988, p. vii). Central to this book is the appealing idea that designers should focus upon users' perceptions, conventions and errors rather than valuing aesthetics over usability (1988, pp. 150-151).

He advocates two things (p. 12) as 'fundamental principles of designing for people: (1) provide a good conceptual model and (2) make things visible'. Given this centrality of visibility to his thesis (p. 4), it is unsurprising that Norman adopted ideas he had discussed with Gibson when both scholars were at La Jolla (Norman, 1999).

Affordances provide strong clues to the operations of things. Plates are for pushing. Knobs are for turning. ... When affordances are taken advantage of, the user knows what to do just by looking: no picture, label or instruction is required. (Norman, 1999, p. 9)

However, Norman disagreed with Gibson about whether the mind processed or simply 'picked up' information (Norman, 1999; see also Norman, 1988, p. 219; 1969). Consequently, he introduced a subtle but important change into the way he used the term: 'The term *affordance* refers to the perceived and actual properties of the thing, primarily those fundamental properties that determine just how the thing could possibly be used' (Norman, 1988, p. 9).

He later expanded on the difference he intended:

POET was about 'perceived affordance'. When I get around to revising POET, I will make a global change, replacing all instances of the word 'affordance' with the phrase 'perceived affordance'. The designer cares more about what actions the user perceives to be possible than what is true. (Norman, 1999, p. 21)

This shift solves the problem of specifying what a thing affords. However, his user-centred alternative creates other problems. Firstly, 'affordance' becomes redundant as an analytic concept. Norman clarified: 'There can be both real and perceived affordances, and the two sets need not be the same' (Norman, 1999, p. 24).

The implication of this, however, is that 'real' affordances are unknowable. If any given set of perceived affordances may be different to the 'real' one, how can we ever know which set is 'right'? All we have access to is what we can perceive – thus all we can ever access are 'perceived' affordances. Norman does not seem to recognise this problem, let alone resolve it; he sees no inconsistency in his interpretative position and the positivist claim: 'Affordances reflect the possible relationships among actors and objects: they are properties of the world' (Norman, 1999, p. 10).

There are related problems. To explain these, it is necessary to introduce other concepts from POET. Norman differentiates between affordances, constraints and mappings (Norman, 1988, p. 12). Constraints may be physical, semantic, cultural or logical. He proposes that physical constraints can be used to prevent certain operations and that 'desired actions can be made obvious, usually by being especially salient' (p. 84). Oddly, since 'obviousness' implies interpretation, he does not position this as a semantic constraint. Indeed, he has remarkably little to say about these, other than: 'Semantic constraints rely upon our knowledge of the situation and of the world. Such knowledge can be a powerful and important clue' (p. 85).

Arguably, these could be incorporated into the next concept: cultural constraints. These he describes as 'conventions' and reduces to cognitive schemas or frames (pp. 84-85). He recognises them as significant sources of error but has little to say about how to design with or to overcome them. In spite of recognising their utility, he rails against their arbitrariness, since there are all too often situations in which such conventions fail to hold (e.g. p. 168).

Interestingly, as an example, he states, 'one cultural convention is that signs are meant to be read'. This is important, given his commitment to perceived affordances. Gibson's affordances are supposed to cross the subjective/objective binary, but this contingency is a concession that interpretation (even at the level of engagement with a design) is learnt, not something 'in' the artefact. This contingency arising from interpretation is compounded where he discusses how people 'explain away' errors or ignore their beliefs due to peer pressure: 'Everyone forms theories (mental models) to explain what they have observed. ... People are free to let their imaginations run free as long as the mental models they develop account for the facts as they perceive them' (p. 39). They have no way of telling which account is 'true' – even hindsight is merely justifying through post hoc rationalisation (pp. 128-129).

What is left, then, of the supposed objectivity of affordances? All we can work with is socialisation and learning. Even Norman's discussion of logical constraints rapidly returns to 'natural mappings' that are actually conventional. Interestingly, that 'naturalisation' is central to Norman's thesis is illuminating. To position something as natural is to render it invisible, unquestionable; to imply that it could not be otherwise (Potter & Wetherell, 1987). Consider, then: 'The vertical plate and supporting pillars are *natural* signals, *naturally* interpreted, without any need to be conscious of them. I call the use of natural signals *natural design* and elaborate on the approach throughout this book' (Norman, 1988, p. 4). He goes on to claim that car design is

‘natural’ (p. 22), revealing that he means something specific by this: ‘Natural mapping, by which I mean taking advantage of physical analogies and cultural standards, leads to immediate understanding’ (p. 23).

Analogies (even physical ones) are clearly interpretative, the product of socialisation, leaving ‘nature’ as a symptom of culture. He further differentiates between ‘natural’ (physical) properties and ‘artificial’ (cultural) conventions, criticising the latter for being difficult to learn, since rote learning (‘the bane of modern existence’, p. 68) gives no hint about the underlying conceptual model. He does, however, recognise such ‘arbitrary’ conventions as a useful last resort: ‘When something can’t be designed without arbitrary mappings and difficulties, there is one last route: standardize. ... The nice thing about standardization is that no matter how arbitrary the standardized mechanism, it has to be learned only once’ (1988, p. 200). This all defines ‘good’ design culturally, not in terms of properties of the world.

Furthermore, he explains how a ‘good’ mental model ‘transforms an apparently arbitrary task into a natural one’ (1988, p. 69). This suggests that personal meaning is key, for Norman, and that it can even be created for arbitrary designs. The quality of design thus lies not in its substance, but in the ability of the user to read it – or to imagine how to read it. This is made clearer in a section where he discusses the role of three conceptual models in design:

Three different aspects of mental models must be distinguished: the *design model*, the *user’s model* and the *system image*. ... The design model is the conceptualization that the designer has in mind. The user’s model is what the user develops to explain the operation of the system. Ideally, the user’s model and the design model are equivalent. However, the user and designer communicate only through the system itself: its physical appearance, its operation, the way it responds, and the manuals and instructions that accompany it. Thus the *system image* is critical: the designer must ensure that everything about the product is consistent with and exemplifies the operation of the proper conceptual model. (1988, pp. 189-190)

This passage is useful in re-casting design as communication. It highlights the importance of making intentions – meaning – clear; it re-positions the user as a reader and the designer as an author, with design as a form of semiotics. The problems of design become social ones of sharing meanings. This sense of design is echoed throughout the book (e.g. 1988, pp. 2, 198). Thus to take advantage of the ‘affordances’ of things, users must learn to interpret their design, based either on past experience or on information presented to us (1988, p. 82).

Norman’s assertion that ‘affordances...convey messages about their possible uses, actions, and functions’ is clearly at odds with Gibson’s position that the world does not communicate with us (Norman, 1988, p. 242). Indeed, so little of Gibson’s intended sense of the word remains that the appropriateness of its use must be questioned. Moreover, as McGrenere & Ho point out (2000, p. 3), Norman has conflated affordances with the information specifying affordance; as argued earlier, given the problematic (and unknowable) nature of affordances per se, either falling back on species-level generalisations or a retreat to learnt responses seem to be the only viable courses of action.

Norman thus leaves himself in an untenable position. If affordances are subjective and interpretative, they cannot form an unambiguous basis for design. He is left in the limited position of only considering ‘normal’ users – those who have already learnt simple responses to things. (Gibson’s position is similar, in that species-level similarities are frequently assumed.) He also assumes, patronisingly at times, that complex things may be too hard to learn (e.g. an application sounding remarkably like Microsoft Word being criticised because ‘there is no way that a program can remain usable and understandable by the time it has all of those special-purpose features’ [1988, pp. 172-173; cf. p. 213]). Norman does recognise that people’s learning is heterogeneous, differentiating (for example) between designers and ‘typical users’ and discussing deterioration of ability with age (1988, pp. 151, 155, 162-163).

Thus, focusing on the role of learning and cultural contexts offers a new way to understand the point of POET. If ‘natural’ is understood as ‘taken-for-granted culturally’, we end with a position that is consistent and educationally relevant – but nothing to do with Gibson’s affordances. Norman may have introduced affordances to design, but what remains valuable in his position is the role of cultural conventions, which he was so critical of.

Affordances in Human–Computer Interaction

Given Norman's explicit discussion of computers and design, it is hardly surprising that his version of 'affordance' was taken up by human–computer interaction (HCI) researchers. McGrenere & Ho (2000) have provided a review of uses of the term, identifying several parallel conventions:

- Gibson's notion of an action possibility or offering;
- Norman's notion of a perceived suggestion;
- As an interface object; and
- In unclear or inconsistent ways.

They suggest 'affordance' is often understood as 'the design aspect of an object which suggests how the object should be used' (2000, p. 1) because Norman collapsed the utility of an object with the perception of that object's utility (2000, p. 3).

McGrenere & Ho advocate a return to a developed version of Gibson's formulation. Two developments are proposed (2000, p. 2): analysis of complex affordances that arise from hierarchies of simpler affordances, and also analysis of degrees of affordance (focusing on minimising effort), rather than a binary scheme. They also distinguish the utility of an object (Gibson's sense) from its usability (Norman's sense) in a way that is productive for designers. Similarly, Torenvliet (2003, p. 14) discusses how affordance in Gibson's sense can be useful 'because it forces us to think more than screen deep', considering how 'clicking' an icon actually entails a physical mouse button press leading to a series of state changes within the computer. (He also criticises Norman's influence, leading HCI researchers to only consider 'perceived' affordances, effectively reversing the definition of the term so that it indicates two contrary meanings.)

Albrechtsen et al (2001), who also reviewed this field, argue that both Norman and Gibson are 'considering affordances as more or less static surface phenomena' (p. 10). They point out that neither really address high-level cognitive issues such as culture, language or knowledge (p. 15); they thus consider 'higher order affordances', linked to Wittgenstein's idea of language games. They do highlight points of continuity with Gibson, such as the focus on relationships, rather than tool or person, although their situated analysis arguably sidesteps the problems of generalisation faced by Gibson. They take an interest in tools as 'extensions of the body' (in Gibson's terms), drawing an analogy with the functional organs described by Leont'ev. Perhaps this consonance is partly because they share a positivistic foundation with Gibson, discussing (for example) the properties of objects or the way that displays present (rather than construct) the 'states of the system and the environment' (p. 26).

Their position is, however, quite a 'development' – they study mediated understanding (contrasted with unmediated direct perception), deny the timelessness of affordances (posing instead an historical analysis), are less concerned with 'natural' affordances (focusing on designed artefacts) and argue that affordances are contingent on the situation, actors and purposes (not essential environmental properties). They also, somewhat dubiously, talk about direct perception of 'the deep structure of the work', asserting (presumably metaphorically) that social structures provide invariants that specify affordances of the work domain. This radically redefines affordances as 'cues for action relevance' (p. 27; making them firmly interpretative) or even as 'social intentions' (p. 30). Again, although their position is interesting and addresses many of the problems identified above, this is achieved by turning Gibson's concept of affordances into re-branded Activity Theory. Quite why the re-branding is necessary, when Activity Theory already seems adequate, is unclear.

Affordances and Learning with Technology

The overlap between HCI and e-learning has led to affordances being taken up by researchers in this area, too. Like HCI, use of the term 'affordance' is variable. For example, Laurillard et al (2000) assert: 'Examples for education would be: A large lecture affords listening. A small group also affords preparing to speak. In each case the features as perceived by an observer create the possibility for a certain kind of behaviour' (p. 3).

The first example suggests Gibson's concept. The second does not involve action as understood from an ecological perspective. (Preparing to speak, rather than speaking, is arguably a

private, 'internal' process, failing to meet the requirement of affordances as related to animal–environment pairings.) The explanation, focusing on what is perceived, suggests Norman's use of the term. Although it might be possible to interpret all of these as perceived affordances, making no claims about what is 'offered', it is Gibson, not Norman, who is referenced.

Particularly useful in illustrating this tradition are the discursive articles provoked by Conole & Dyke's proposal to use affordances as the basis for research (2004a). Their definition is derived from Norman by way of Salomon. However, although they assert (2004b, p. 301) that by this they mean to focus on "'possible" use', they discuss a negative sense in which affordances of objects 'often restrict their accessibility to users' (2004a, p. 115). Equally, although they clarify that they are interested in the relationship between technological infrastructure and people's use of those technologies, they are still concerned with the 'properties of a thing' (albeit those that are perceived as well as those it possesses), once again committing to a positivist, essentialist position. (Even in the subsequent clarification, qualities such as 'speed of change' are described as 'an inherent and important characteristic of technologies' – 2004b, p. 304.) This position is critiqued by Boyle & Cook (2004), who point out the discrepancy between the espoused social basis of Conole & Dyke's analysis and Gibson's explicitly unsocial, non-constructivist position.

Conole & Dyke then draw on 'relevant social theory and critique' to create 'a list of the potential themes and commonalities ... distilled into a taxonomy of ICT [information and communication technology] affordances' (2004a, p. 116). This list groups together qualities attributable to the technology (e.g. multimodality and non-linearity), to its user (e.g. reflection) and to their mutual relationship (e.g. immediacy). Arguably, such a list does not conform neatly to Gibson's, Norman's or McGrenere & Ho's formulation of affordance. Some elements seem consistent with the essentialised, positivist origins of affordance. Others seem entirely unrelated – reflection, for example, would be denounced by Gibson. Moreover, the idea that reflection might be a response to an offering by technology (implying some causal link) rather than an act of personal agency seems odd. Additionally, Conole & Dyke assert that they 'are also interested in exploring the creative and innovative way that people respond to technologies and perhaps adapt them for use in unforeseen circumstances' (2004b, p. 301), rather than just considering designers' intentions. This recognises earlier critiques of the predictive value of Gibson's affordances but does not resolve them; there is still no way of identifying these possibilities or specifying what they consist of.

Indeed, there seems to be no unifying concept behind the list; elements may emerge from the literature, but the notion of 'affordance' seems ill-suited to legitimating this conglomeration of claims about perceptions, actions and characteristics. Something much broader is required. Substituting the word 'claims' for 'affordances', for example, provides a more plausible framework with no loss of the central message and no diminution of utility to practitioners. For example:

Many of the [claims] outlined above are known intuitively, but the development of an explicit taxonomy provides a basis for discussion, critique, and further refinement. This may involve questions such as:

- Are these the only [claims]? ...
- Can particular [claims] be mapped to particular types of technologies? (Conole & Dyke, 2004a, p. 121)

Repositioning the taxonomy as a socially constructed account bypasses the ontological confusion and inconsistency inherent in the current formulation – although it does not confer the status that Gibson's oft-cited, 25-year old text might. It also helps explain Boyle & Cook's criticism (2004, p. 297) that the paper contains 'a certain amount of hopeful expectation that affordances and abilities will simply emerge', since these themes *are* emergent, having been synthesised from research narratives.

Boyle & Cook suggest alternatives. One is to extend the 'ecological' approach underlying Gibson's thesis so as to focus upon how technology has changed the 'habitat' in which we live. This suggestion echoes the 'information ecologies' proposed by Nardi & O'Day (1999), but ignores the issues facing Gibson's formulation of 'affordance' and also its primitive and 'strikingly constant' (Gibson, 1979, p. 19) scope. Gibson does not consider the idea that an animal might develop its 'niche' (the collection of affordances that make a particular ecological setting a suitable habitat). Nonetheless, if the other issues in Gibson's thesis (especially the attitude towards the mind) were

also addressed, this proposal may be worth pursuing – although it remains interpretative, rather than helping to identify the possibilities for action ‘afforded’ by technology that prompted interest in the term.

The second suggested alternative focuses on ‘social affordances’, which are interpreted as properties incorporated into computer-supported collaborative learning environments that ‘act as social facilitators’ (Boyle & Cook, 2004, p. 298). ‘When they are perceptible, they invite the learner to act in accordance with the perceived affordance, i.e., start a task or a non-task related interaction of communication’ (Kreijns et al, cited in Boyle & Cook, 2004, p. 298). Examples include tools for monitoring communal activity. Such tools might be useful, but they miss the point of affordances – they extend what is noted (by supporting the perceptual process with more visible signals) but they do not ‘offer’ new possibilities for action; they are simply intended to encourage particular extant social activities. Here, the extension of Norman’s use of the term becomes particularly misleading, confusing usability and utility.

Discussion

Use of the term ‘affordance’ diverged after Norman appropriated it. One line of development is consistent with Gibson’s original position, although arguably these refinements have not countered problems with its relevance to learning. This tradition is positivistic and essentialist, at odds with contemporary educational thought. This is, perhaps, inevitable since Gibson seems preoccupied with ontology, not learning, and explicitly sought to rule out the “‘subjective” world,’ or the world of “‘consciousness”” (Gibson, 1979, p. 129). Arguably, it would be inappropriate to advocate a return to this position in educational research.

Jones (2005) argues that it is possible to overcome these difficulties by emphasising the non-dualist, realist and relational nature of affordances. However, this position does not address either of the two main critiques offered. No ontological explanation is provided of what this relationship between animal and environment consists of. (Is it historical, imagined or something else? Even if it is one of these, the problem of whether this resolution has implications for predictive analysis still remains – this prescription of possible interactions being the hope that gives the concept its analytic appeal.) Second, the assumption that a totalising ‘view from nowhere’ can be inferred from specific perceived instances fails to account for the idea that affordances might exist but never be perceived, and is also an epistemologically suspect inferential move (cf. Albrechtsen et al, 2001).

The second major tradition builds on Norman, using the term in a looser sense. The analysis above suggests that Norman’s position may have consistency, but not for the reasons he proposed. If design is reinterpreted as a form of cultural literacy this position addresses the predictive requirements that are sought, but affordances have no analytic role to play in this position, and indeed in their original sense are inconsistent with it (cf. Albrechtsen et al, 2001). To adopt this position, arguably, the term ‘affordance’ should be abandoned as being misleading since its original connotations cannot be left behind.

A third position is suggested by Boyle & Cook (2004), which might involve developing the ecological approach further. It may be that Nardi & O’Day’s (1999) work on information ecologies contributes to this, but since affordance is not a major part of that work, this approach will not be pursued further here.

The problem with these divergent lines of development is that ‘just as printing new money devalues existing money, the more new definitions the term affordance gains, the less value any one of them has’ (Torenvliet, 2003). Where, then, does this leave researchers and designers?

I have No Affordances, Yet I Must Design

Arguably, researchers and designers persist in using the term affordance because of the desire to speak about technology in a general enough way that it can be theorised in its own right. It also promises an appealing control over users – if affordances permit some actions and constrain others, users are more likely to behave as expected. There is thus a tension between recognising the complexity of the concept and having a language simple enough to be useful.

Does it matter, however, if the term is used inconsistently, if it is productive? Arguably, yes. For researchers, incoherence in terminology will provide a false impression of research coherence, hiding (not resolving) problems. For designers, the concept might prompt reflection, but if that reflection underplays the role of the user (Norman's critique of Gibson) or of learnt cultural conventions (the critique of Norman advanced here), then increasing its quantity, rather than its quality, is a disservice to users. Some coherent, plausible alternative is needed.

There are at least two methodologically and epistemologically coherent ways in which this tension can be met. The first extends the claims-based approach used (not espoused) by Conole & Dyke. Working with accounts of technology (reported experience, research papers and so on.) brings the discussion, epistemologically, to a single level: discourse. These discourses can be identified, genealogies constructed, their use and conflicts studied. Methodologies for this exist, typically drawing on Foucault's work.

However, some may find this retreat from the object (technology) to talk about the object unsatisfying. There may be an alternative, although its viability is yet to be demonstrated.

The analysis above of designed artefact as cultural text suggests that it may be possible to undertake a literary analysis of technology. Studying the stylistics of technology may permit it to be talked about without resorting to a positivistic position. (This perspective is not entirely unprecedented; Albrechtsen et al, 2001, talk about artefacts as 'texts' or 'narratives' arising from research into intuitive interfaces [p. 12].)

Bakhtin's difficulty in studying the novel is analogous in many ways to the situation here:

The study of the novel as a genre is distinguishable by particular difficulties. This is due to the unique nature of the object itself: the novel is the sole genre that continues to develop. ... The forces that define it as a genre are at work before our very eyes: the birth and development of the novel as a genre takes place in the full light of the historical day (Bakhtin, 1987a, p. 3).

There is particular merit, as well as difficulty, in studying such evolving forms. Bakhtin argues (p. 7) that as it develops 'it reflects more deeply, more essentially, more sensitively and rapidly, reality itself in the process of unfolding'. It thus provides an important lens through which to study culture more broadly; in this case, studying technology illustrates issues about education more widely.

This does not require a simple and positivistic position, but sees 'reality' as social and historical (Bakhtin, 1987b, p. 263). Part of this position – one that offers interesting but relatively neglected possibilities for the study of technology and education – is the assertion that all language use is ideological (Bakhtin, 1987b, pp. 270-271). Additionally, the novel is presented as reflecting heteroglossic language use. Studying the way in which technologies are brought together to automate practices (such as surveillance, communication and so on) – practices that may or may not cohere successfully – promises to reveal the ideological positioning of particular systems, designs or devices through a reading of the elements that they consist of. This would allow technology to be described in a way that recognises its social and historical production, as well as intentionality, without proposing simple causal relationships.

Importantly, Bakhtin positions the ideology of language as dialectic, not transmissive. Speakers can appropriate 'alien' expressions, making them their own (Bakhtin, 1987b, p. 282). This echoes Conole & Dyke's (2004b) desire to recognise creative uses of technology, beyond the intended use. However, it does so analytically, exploring the intentionality of various actors (Bakhtin, 1987b, p. 292). There is no recourse to a speculative set of possibilities, but instead the analysis of concrete instances of expression – something as coherent as the previously proffered analysis of discourses, but maintaining a focus on technology. This resolves the difficulty of specifying the nature of the relationship between technology and people. It does not claim to solve the problems of prediction facing affordances, although it is coherent with an approach that involves studying existing repertoires of use and recognising the potential of analogous or newly imagined applications.

Methods for undertaking such work have yet to be developed, but this analysis suggests that such work may be productive.

Conclusions

This article calls into question whether the concept of ‘affordance’ has any analytic merit for researchers or designers. The critiques advanced here suggest that its value is limited *unless* we are willing to abandon constructivist values in order to explore ‘inherent properties’ in a positivistic sense. The concept *either* allows us to talk about technology as part of the environment (Gibson’s sense), but not about minds; *or* is rendered analytically ineffectual by recognition of the cultural, constructive nature of learning (Norman). The only developments that seem to escape this dilemma (e.g. Albrechtsen et al) do so by keeping the label but effectively abandoning the concept (e.g. by it becoming Activity Theory). On the one hand, we abandon current conceptions of learning; on the other, we effectively abandon affordance – although researchers seem remarkably reluctant to give up the term.

Even in Gibson’s sense the term is problematic. It may be adequate for discussions of species and generalities, but people and learning are beyond it. Furthermore, the need for people to learn in order to take advantage of affordances is unexplained and Gibson fails to provide any analytic way of identifying affordances. His discussions resort to commonsense examples that are not exhaustive or definitive, merely illustrative. This does not give a method for research or design, and led to Norman’s veiled reversion to designing around cultural norms. By conceding that affordances need to be learnt but failing to explain their relationship to ‘attunement’, Gibson leaves himself vulnerable to the same charge that he levelled at cognitive processing theorists (1979, p. 238): there are too many perplexities, and thus this explanation should be abandoned.

His position gives primacy to environment over people and only recognises people as tool using, not tool making; it ignores our agency. It may be more productive for this field to focus not upon the ‘offered’ possibilities but upon what a person *imagines* might be possible – and also upon what they can imagine doing to achieve the same end with some other object. If an object does not allow us to undertake an action, we can find an alternative or make a new tool that does.

Norman’s re-interpretation seeks to recognise some of these issues, but by doing so he renders the notion of ‘real’ (as opposed to ‘perceived’) affordances irrelevant. This has led to the current position in which the term is used in a degenerate way. Since these different traditions cannot be eradicated, this confusion will persist, muddling both research and design; arguably, then, the term ought to be abandoned. Nevertheless, the coherent elements of his position can be reinterpreted in a way that has promise. Using literary techniques to study how people ‘write’ (design, create) and ‘read’ (use and apply) technology may support research that is coherent and socially aware without trivialising the complexity of this field of study by resorting to an essentialist, positivist position. We should just avoid calling what we study ‘affordances’.

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